

## 4.2 Linear Approximation & Differentials

In this section we will go over another important application of differentiation, and we will attach some rigorous meaning to the notion of a differential.

**Motivation:** What is an example of a function that is easy to calculate for some inputs, but quite difficult to compute by hand for other inputs?

*Note: An important question is: How can we tell how far off an approximation with a tangent line will be from the actual value we seek with no computation device? This question is very important in computer science and will be addressed in Calculus II. In Calculus I we are in the unfortunate position of knowing that the tangent line yields approximations, but the only way we can tell how good an approximation is, is to use the calculator.*

The discussion in the motivation leads to the following definition.

**Definition:** The line  $L(x) = f'(a)(x-a) + f(a)$  is called the **linearization** of  $f$  at  $a$ .  
The approximation  $f(x) \approx L(x)$  is called the **linear approximation** of  $f$  near  $a$ .

*Note:* In the motivation, we found that the linearization of  $f(x) = \sqrt{x}$  at 1 was  $L(x) = \frac{1}{2}x + \frac{1}{2}$ .

**Example 1:** Consider the function  $f(x) = x^6$ .

a) Find the linearization of  $f$  at  $x = 2$ .

b) Use your answer from part (a) to approximate the number  $(2.01)^6$  by hand.

c) Use your calculator to find how far off your approximation in part (b) is from the actual value of  $(2.01)^6$

**Example 2:** Use a linear approximation to estimate the value of  $\ln(1.5)$ .

**Definition:** If  $y = f(x)$  where  $f$  is a differentiable function, then

- the **differential**  $dx$  is an independent variable; that is,  $dx$  can be given the value of any real number.
- the **differential**  $dy$  is then defined by the equation  $dy = f'(x)dx$ .

So  $dy$  is a dependent variable that depends on the values of  $x$  and  $dx$ .

*Note: This definition is in agreement with Leibniz notation for the derivative. In particular,  $\frac{dy}{dx} = f'(x)$ .*

**General Idea:**

The geometric meaning of a differential can be seen in the following diagram.

**Example 3:** Suppose that  $f(x) = x^3 - 1$ . Find the values of  $\Delta y$  and  $dy$  (rounded to 4 decimals) if  $x$  changes:

a) from 2 to 2.4.

b) from 2 to 2.05

**Example 4:** Let  $y = \sqrt{x-3}$ .

a) Find the values of  $\Delta y$  and  $dy$  (rounded to 4 decimals) if  $x$  changes from 4 to 7

b) Make a sketch of  $y = \sqrt{x-3}$ , showing the values of  $\Delta y$  and  $dy$ .